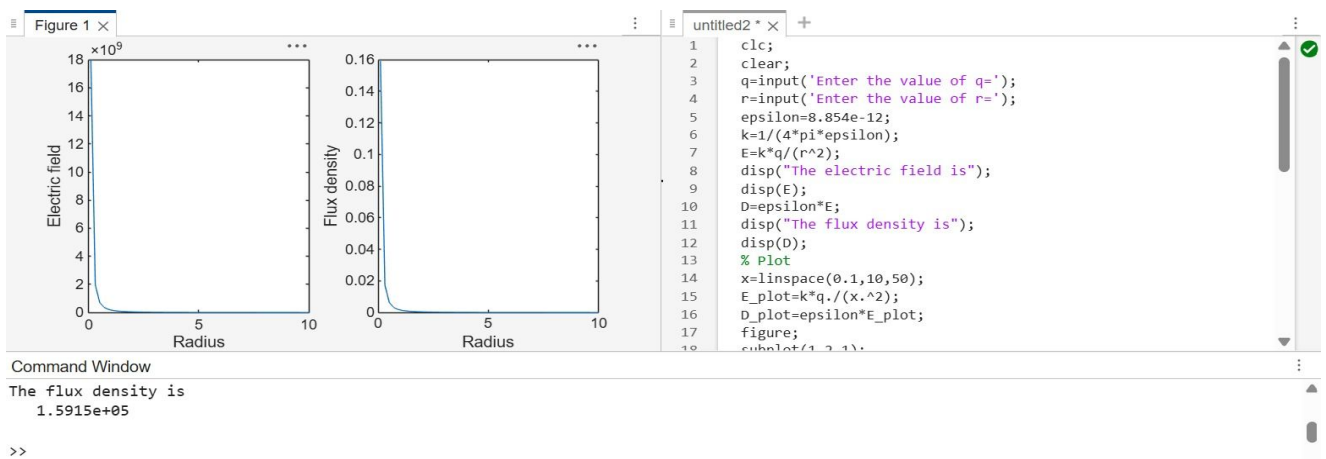
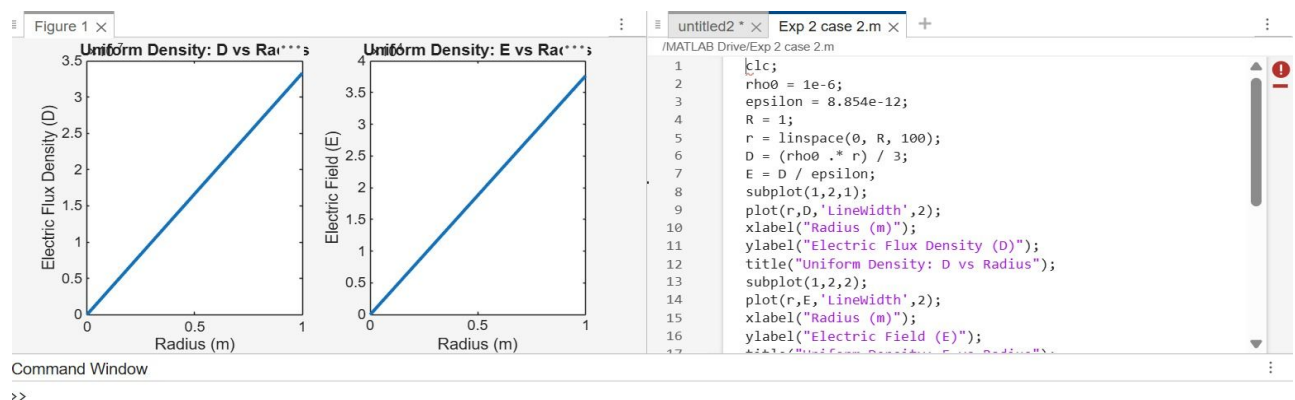


Exp. No: 2 Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius

Test case (i) Determination and plotting of electric field and electric flux density in sphere full of charges with varying radius

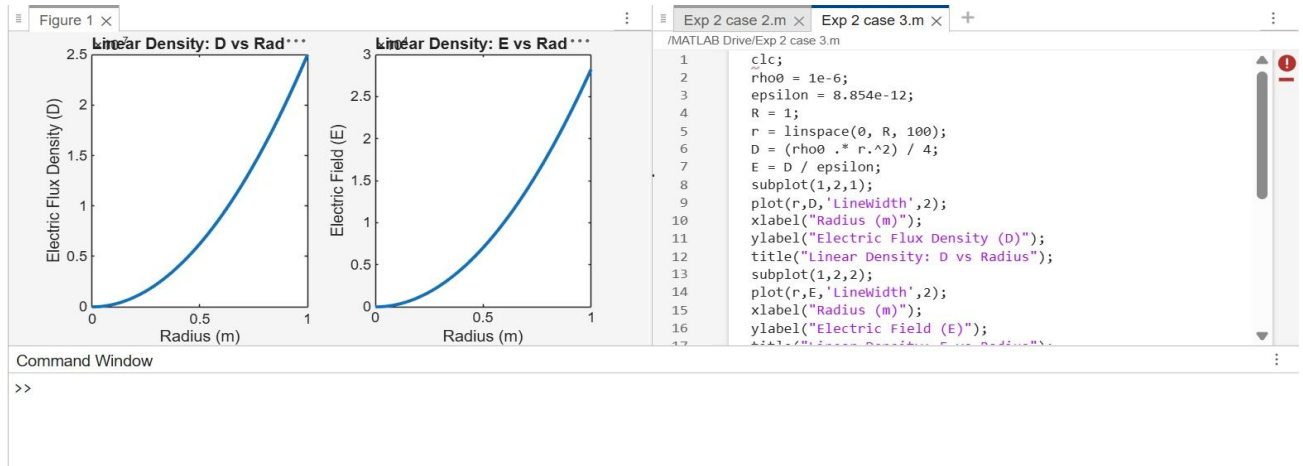


Test case (ii) Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ_0



Test case (iii) For a sphere with a linearly increasing charge density $\rho_v = \rho_0 r$, derive and plot

the electric field (E) and electric flux density (D)



Test case (iv) A sphere of radius $R = 2$ m has a uniform charge density $\rho = 3 \times 10^{-6}$ C/m³.

Calculate and plot electric field inside and outside the sphere

